

Food Delivery Management System

Complete UML Diagram Documentation

This document presents a complete set of UML diagrams covering all levels of the Food Delivery Management System — from requirements (use case) to structure (class), behavior (sequence, activity, state), and architecture (component). The system supports four actors: **Customer**, **Restaurant**, **Delivery Agent**, and **Admin**, enabling end-to-end food ordering, payment, preparation, and delivery tracking.

#	Diagram	UML Level
1	Use Case Diagram	Requirements
2	Class Diagram	Structural
3	Sequence Diagram	Behavioral (Dynamic)
4	Activity Diagram	Behavioral (Dynamic)
5	State Diagram	Behavioral (Dynamic)
6	Component Diagram	Architectural

1. Use Case Diagram

Captures the functional requirements of the system by showing how the four actors interact with the system's use cases.

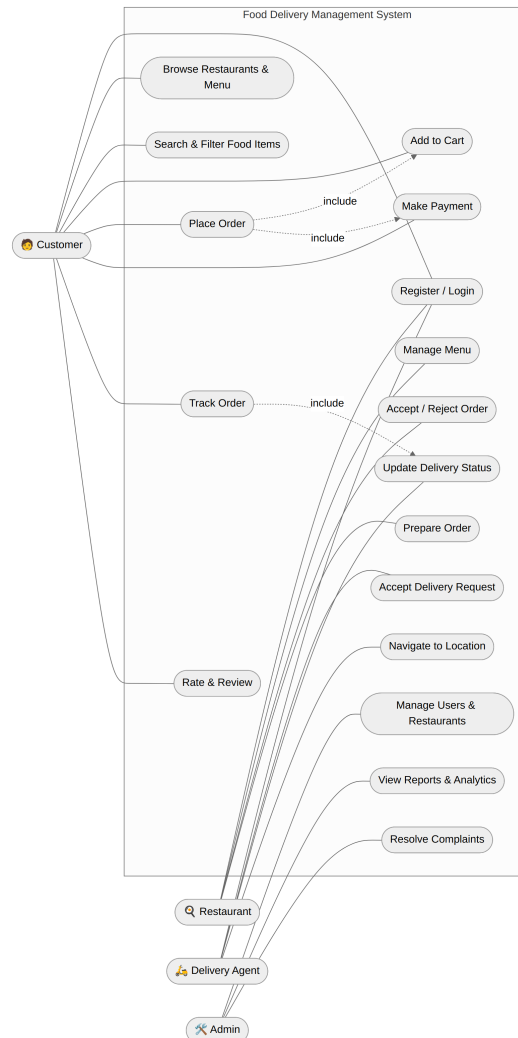


Figure 1: Use Case Diagram

Key Points

- **Customer:** registers/logs in, browses restaurants, places orders, pays, tracks delivery, and reviews.
- **Restaurant:** manages the menu and accepts/rejects and prepares orders.
- **Delivery Agent:** accepts delivery requests, updates status, and navigates to the destination.
- **Admin:** manages users/restaurants, views analytics, and resolves complaints.
- "Place Order" includes "Add to Cart" and "Make Payment"; "Track Order" includes "Update Delivery Status".

2. Class Diagram

Models the static structure of the system: the core entities, their attributes/operations, and the relationships (inheritance, association, composition) between them.

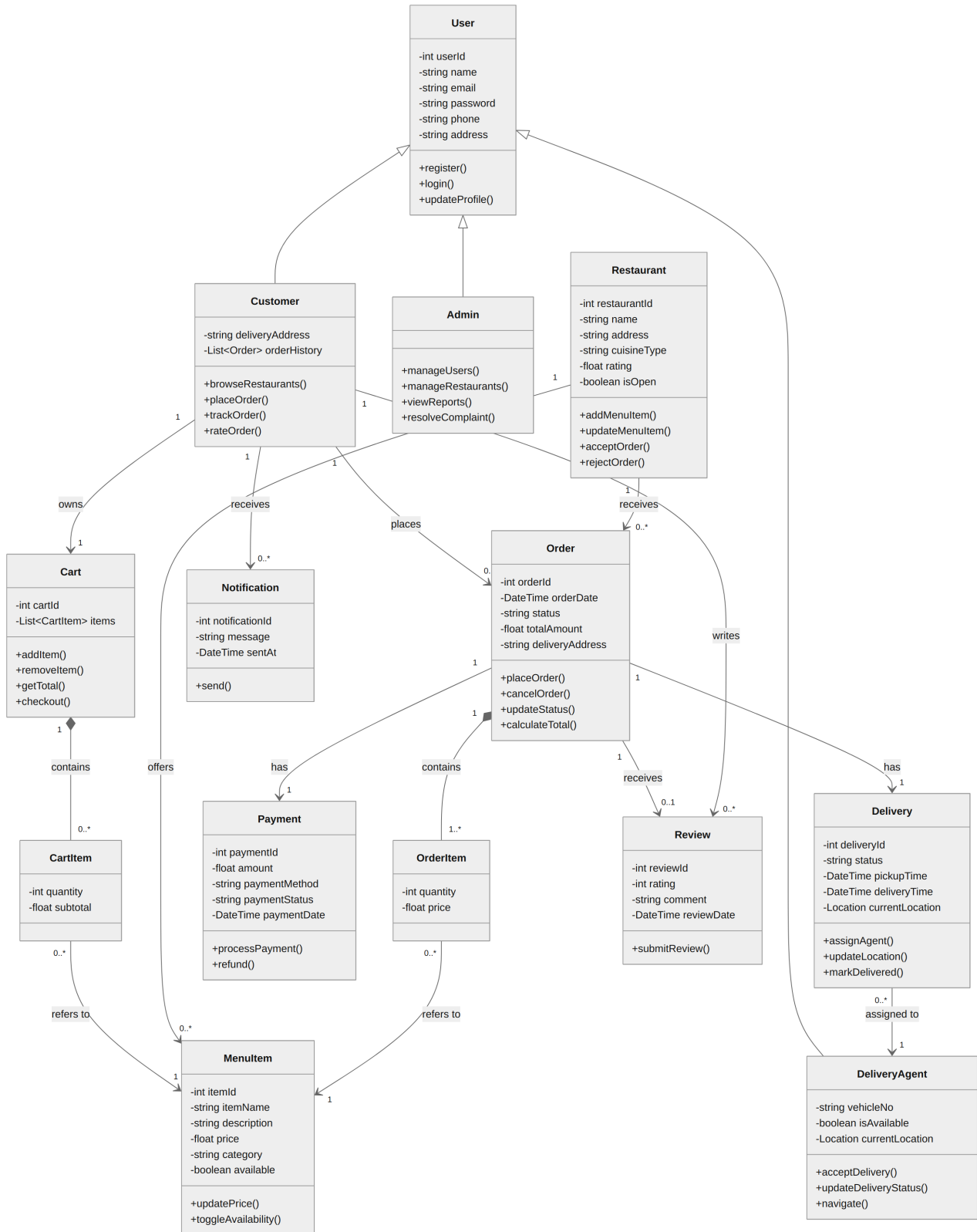


Figure 2: Class Diagram

Key Points

- **User** is a base class specialized into **Customer**, **DeliveryAgent**, and **Admin** (inheritance).
- A **Cart** is composed of **CartItems**, each referencing a **MenuItem**.
- An **Order** is composed of **OrderItems** and is linked one-to-one with **Payment** and **Delivery**.
- A **Restaurant** offers many **MenuItems** and receives many **Orders**.
- **Delivery** is assigned to a **DeliveryAgent**; completed orders may receive a **Review**.

3. Sequence Diagram — Place Order Flow

Shows the time-ordered interaction between the Customer, the App, the Restaurant, the Payment Gateway, and the Delivery Agent for a complete order-to-delivery cycle.

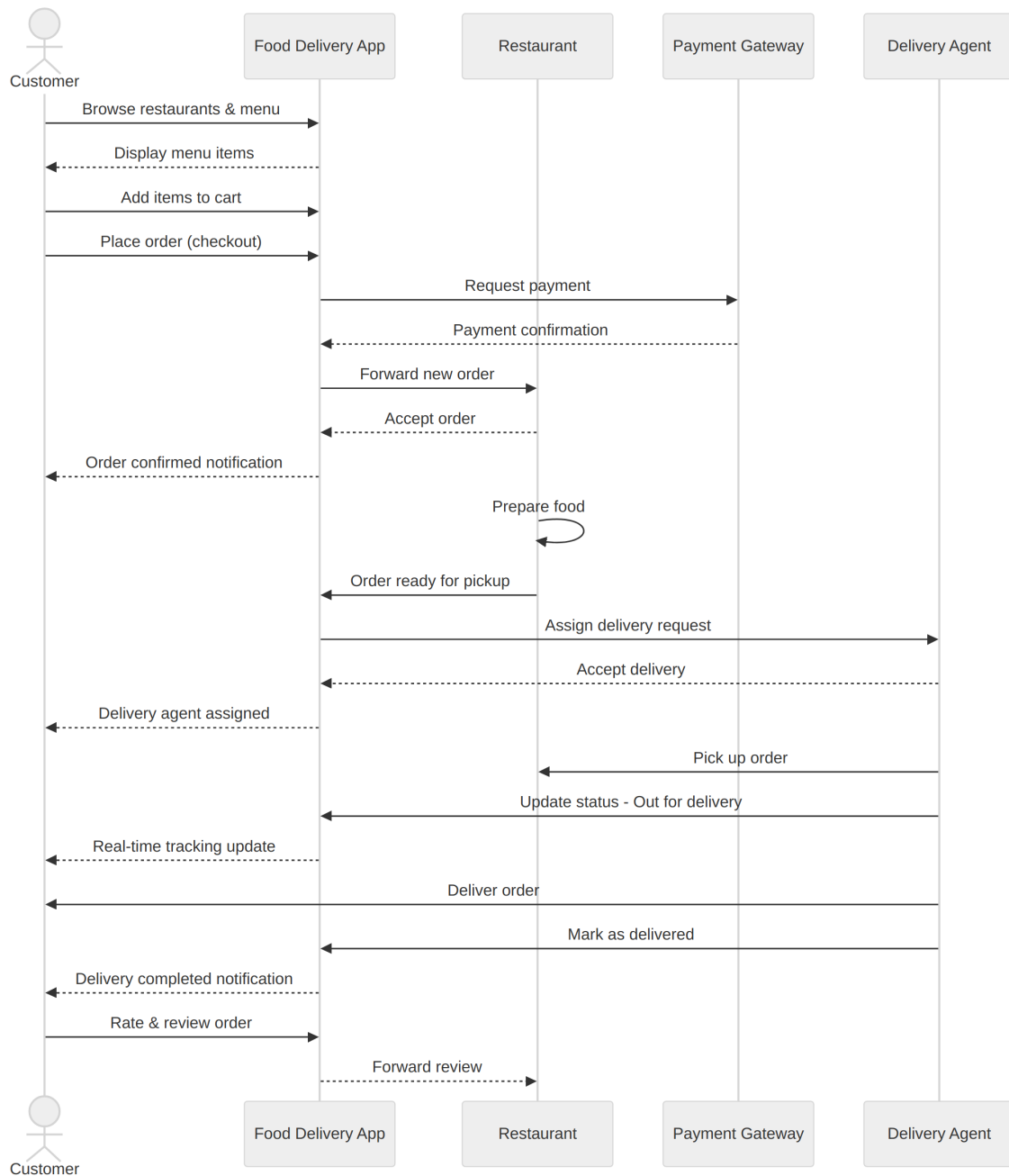


Figure 3: Sequence Diagram — Place Order Flow

Key Points

- The customer browses the menu, builds a cart, and checks out.
- Payment is processed before the order is forwarded to the restaurant for acceptance.
- Once ready, the app assigns a delivery agent who picks up and delivers the order.
- Status updates flow back to the customer in real time; a review closes the cycle.

4. Activity Diagram — Order Processing Workflow

Illustrates the step-by-step workflow and decision points (validation, payment, restaurant acceptance, agent assignment) from browsing to order completion.

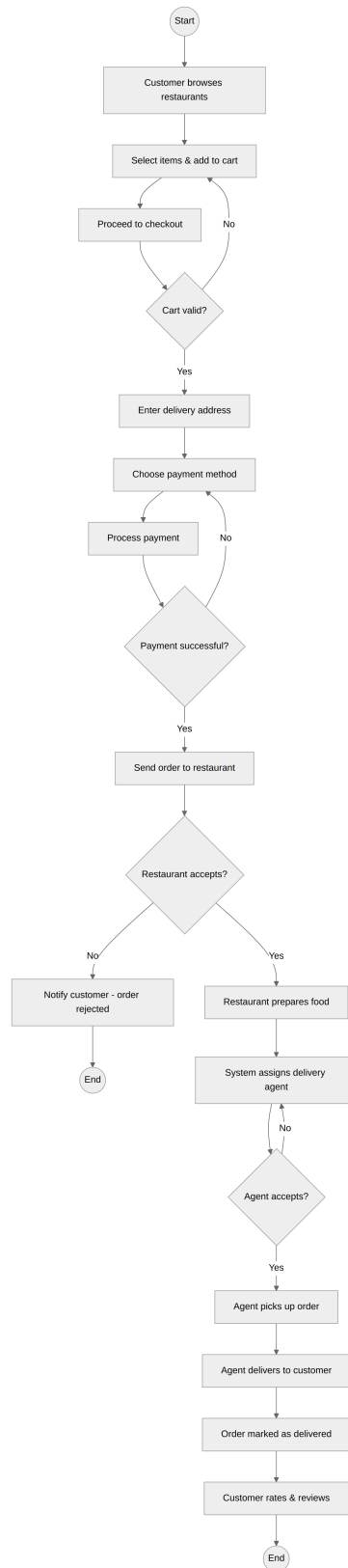


Figure 4: Activity Diagram — Order Processing Workflow

Key Points

- Decision points include cart validation, payment success, and restaurant acceptance.
- A failed payment loops back to payment method selection; a rejected order ends the flow with a notification.
- Delivery agent assignment retries automatically until an agent accepts.
- The flow ends after delivery confirmation and the customer's rating/review.

5. State Diagram — Order Lifecycle

Describes the states an **Order** object moves through, from creation to final resolution (delivered or cancelled), and the events that trigger transitions.

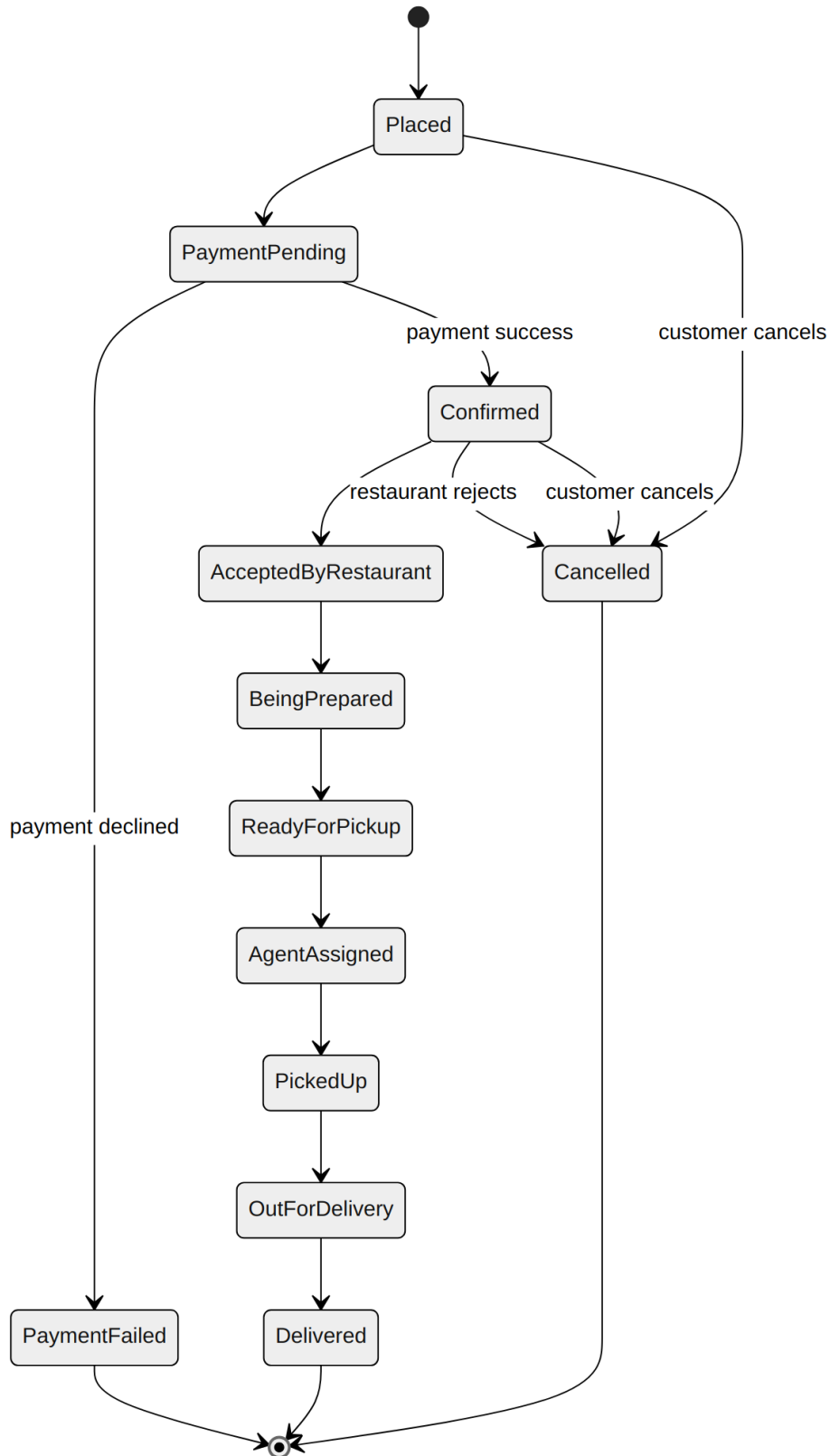


Figure 5: State Diagram — Order Lifecycle

Key Points

- An order can be cancelled by the customer while it is Placed or Confirmed.
- Payment failure returns the order to a terminal Payment Failed state.
- Restaurant rejection transitions a Confirmed order directly to Cancelled.
- The happy path moves through Accepted → Being Prepared → Ready → Picked Up → Out for Delivery → Delivered.

6. Component Diagram — System Architecture

Presents the high-level architecture: client applications, the API gateway, backend microservices, data stores, and external integrations.

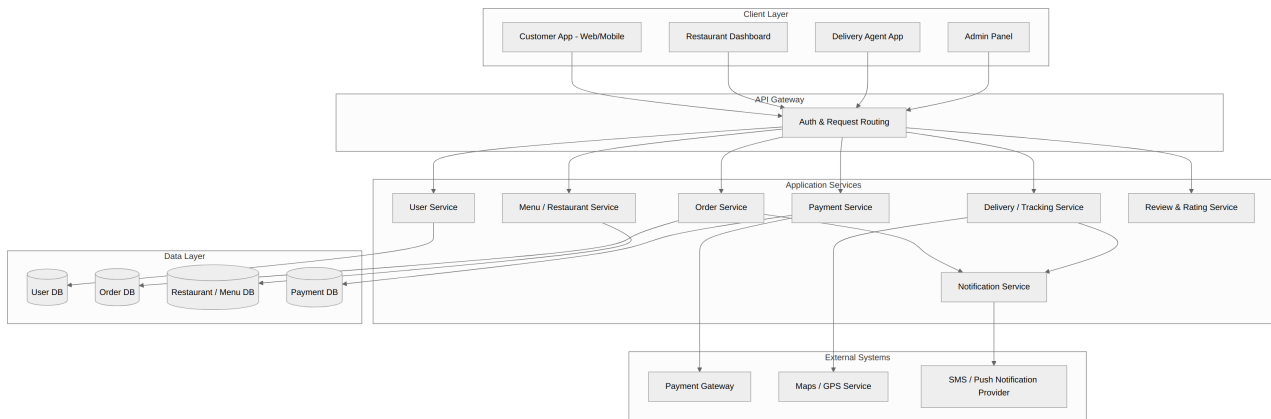


Figure 6: Component Diagram — System Architecture

Key Points

- Four client front-ends (Customer, Restaurant, Delivery, Admin) talk to a single API Gateway.
- Core services — User, Order, Menu, Payment, Delivery, Notification, Review — are independently deployable.
- Each service owns its own data store, following a database-per-service pattern.
- External systems integrated: Payment Gateway, Maps/GPS, and SMS/Push notification providers.